Automated Aeroponics Laboratory System

(AALS)

a Sabetpour Wingate Technologies product

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INTRODUCTION

Aeroponics is a process initiating plant growth without soil. Roots are suspended in an environment and fed nutrients via high pressure misting systems. The benefit to growing plants using aeroponics is a substantially higher rate of yield with minimal utilization of resources such as water, time, and space. Automated Aeroponics Laboratory system, an innovative product from Sabetpour Wingate technologies, will implement a fully automated aeroponics agricultural application using a system-side hardware module comprising an Arduino Due board. The success criteria for this project is dependent on ideal temperature, optimal pH, stable electrical conductivity (EC), oxidation-reduction potential (ORP), and dissolved oxygen (DO). Overall success will be determined by observations of accelerated growth rate compared against accepted timings per plant species.

Supporting Documents

Automated Aeroponics Laboratory System Software Requirements Specification (AALS SRS)

Functional Requirements Specification - see appendix, See also AALS SRS

User Requirements Specification - see appendix, see also AALS SRS

Coding Guidelines and Standards Specification - see Appendix

Validation Plan - see Appendix

ROLES AND RESPONSIBILITIES

- Business Director: Fran Sabetpour

- Development Lead: Anthem Wingate
  - Selects requirements for each sprint
- Requirements Resource: Fran Sabetpour
  - Identifies requirements to implement
- Architect: Anthem Wingate
  - Identify specific functional criteria of the system from a modular perspective
- Designer: Anthem Wingate
  - Identify overall functional purpose of the system
- Buildmeister: Fran Sabetpour
  - Oversee continuous integration and version control
- Developers: Anthem Wingate
  - Implement functional requirements
- Test Lead: Fran Sabetpour
  - Identify and oversee validation plan
- Testers: Anthem Wingate
  - Implement validation plan
- Documentation: Anthem Wingate
  - Create development plan
- Documentation Editor: Fran Sabetpour
  - Edit development plan and supplementary documents
- Risk Management: Fran Sabetpour
  - Identify specifications for system reliability, safety and security
- System Administrator: Anthem Wingate
  - Implement specifications for system reliability, safety and security

As yet unidentified roles

- User advocate: TBD
- Modification Request Board: TBD

- Customer Representative: TBD
- Customer responsible for acceptance testing: TBD

METHOD

- Software:
  - Language(s)
    - C++ 11
    - Java SE 14
  - Operating System(s) with release number
    - Arduino 1.6.8
    - Windows Server 2019
    - Antman 3.3.0 Hypervisor
  - Integrated Development Environments
    - Arduino
    - IntelliJ
    - Visual Studio Code
  - Software packages/libraries used with release/version number
    - Arduino WiFiNina Library
    - Arduino SD Library
    - Additional Java libraries to be determined
    - SLIMS 6.5
    - MariaDB
    - Java OpenJDK8
    - Apache Tomcat 7
  - Code conventions outlined in the Coding Guidelines and Standards Specification - see Appendix
- Hardware:
  - Development Environment
    - Arduino Web Editor Create.arduino.cc
    - Github
    - Travis CI
    - CodeCov.io
    - PostMan
  - Test/Target/Deployment Hardware
    - Antsle XD Ultra VM Server
    - Arduino Due board computer
    - Arduino MKR MEM Shield
    - Arduino MKR ENV Shield
    - Arduino MKR Connector Carrier

        - Arduino MKR WiFi 1010
        - Atlas Scientific Dissolved Oxygen DO Probe
        - Atlas Scientific pH Probe
        - Atlas Scientific Oxidation Reduction Potential Probe
        - Atlas Scientific Electroconductivity Probe

- Back up plan (individual and project)
    - TBD
- Review Process:
    - Informal usability and Code Reviews will be conducted by the testers as part of the validation plan. Code readings will be implemented every spring by the developers prior to build.

- Build Plan:
    - Github, Travis CI, and Codecov.io will be used to implement continuous integration of code for builds and unit testing.
    - Regularity of the builds – a new build will be triggered automatically with every pull request
    - Deadlines for the builds – end of spring builds are due at the end of the day on the concluding date of the sprint.
    - Regression test process – see test plan

- Modification Request Process:
    - MR tool - Github pull requests must be approved by each developer

Virtual and Real Work Space

Google Docs is the repository for project documentation. Pair Programming is implemented using IntelliJ Code withMe and Visual Studio Code Live Share. Conferences are held on Microsoft Teams.

COMMUNICATION PLAN

"Heartbeat" meetings

Occurs on Monday via Slack to provide an update on new code, pull requests, builds, completed requirements, or impromptu changes to the burndown chart or plan for the sprint.

Status meetings

Occurs on Tuesdays, for five minutes at the beginning of the SSW690A class time.

Issues meetings

Occurs as needed via Slack if a build failure occurs. Length determined by time to resolution

TIMELINE AND MILESTONES

• Week of Sept 29th – description of first demo
    WiFi Implementation - Fran Sabetbour
    Data Logger for sensors - Fran Sabetpour
    DataStorage - Anthem Wingate
    Pump and Light Automation - Anthem Wingate
• Week of Oct 6th first demo, description of second demo
    Refactor
• Week of Oct 20th second demo, description of third demo
    Refactor
• Week of Nov 3rd third demo, description of fourth demo
    Flat File Transfer Java Plugin for SLIMS integration (Devise back up plan in case running low on time)
• Week of Nov 17th fourth demo, description of final product
    Refactor
• Week of Dec 8th final product
    Refactor

TESTING POLICY/PLAN

Continuous testing implemented per Validation Plan. Unit tests are run with each build via Travis CI. Usability testing is conducted at the end of each sprint.

RISKS

Some risks include not having enough nutrients. If there are not enough nutrients, then the plant would not be able to grow. Aeroponics is also highly dependent on the system. If the system consisting of the pumps, sprinkles and timers are not functioning properly, then the system would not be able to produce healthy plants. Another risk is through bugs in the code. If there are bugs in the code, then the system would not function properly such as providing logs for the temperature, pH, EC, ORP, and DO. A bug in the code could also result in other situations such as not having the aeroponics system to be fully automated. The best way to keep track and monitor is by observing the behavior of the aeroponics system. Results could even be obtained manually such as the temperature and this can then be compared to the log. If there are no logs, the code needs to be reviewed in order to see where the problem is and how it can be resolved.

ASSUMPTIONS

Automated Aeroponics Laboratory System (AALS) is a product of Sabetpour Wingate Technologies (SWT). SWT is staffed by co-executive officers Fran Sabetpour and Anthem Wingate.

DISTRIBUTION LIST

Dr. Gregg Vesonder, Industry Professor and Director, Research, Systems and Software Division

MORE OPTIONAL SECTIONS:

Worry beads

Backup Plan

IQ design

OQ design

UAT script design

Documentation Plan

See Appendix

Build Plan When builds and testing become complex, this might be a separate section or point to a separate document.

Continuous integration implemented via Travis CI with every Git Pull request.

Appendix

I. **Functional Requirements Specification**
II. **User Requirements Specification**
III. **Code Guidelines and Standards**
IV. **Validation Plan**

Functional Requirements Specification

The pump and lights shall be automated through Arduino.

The system shall log data collected by the sensor array.

The system shall have its database to be integrated with SLIMS.

WiFi Implementation

The system shall have WiFi implementation for Arduino.

Data Logger for sensors

The system shall log data obtained from the sensor array.

DataStorage

The system shall store the data of each automation activity that takes place as a flat file.

The system shall read data from Sensor Array

- MKR MEM Shield shall collect data transferred at the pin connection and write to a text file

The system shall write to SD Card

- Text file shall be named with time in hours_ddmmmyyyy_method of collection
    - Example 1: 2030_15sep2020_ph
    - Example 2: 0630_02dec2020_orp
    - Example 3: 1051_21oct2020_ec
    - Example 4: 1302_06nov2020_d0

Pump and Light Automation

- The system shall regulate power to pump timer
    - Pump on for 0.5 seconds every 2 minutes for 16 hours
- The system shall regulate power to lights timer
    - Lights on for 16 hours

Flat File Transfer Java Plugin for SLIMS integration

- The System shall transfer flat files to SLIMS via Slimsgate API plugin
- The Slimsgate shall read the text file into SLIMS proprietary database.

User Requirements Specification

The user shall be able to run a calibration script for the sensor array to include steps to calibrate temperature, pH, EC, ORP, and DO.

The user shall be able to view a continuous log for data collected by the sensor array to include readings of temperature, pH, EC, ORP, and DO.

The user shall be able to access a menu to navigate to the following options:

- Dashboard
- Data Log
- Calibration

The user shall be able to access the logs within 5 seconds.

Replaceable system parts that fail shall be installed in 2 days or less to minimize disruption of production.

Coding Guidelines and Standards Specification

Coding Guidelines and Standards for Java will be utilized from the following resource:

https://google.github.io/styleguide/javaguide.html

Coding Guidelines and Standards for C++ will be utilized from the following resource:

https://github.com/lefticus/cppbestpractices/blob/master/00-Table_of_Contents.md

Validation Plan for Automated Aeroponics Laboratory System

Introduction

AALS is intended to automate Aeroponics solutions for small scale applications. The goal of this validation plan is to bring AALS to production within the December 8, 2020 timeline. This document details the steps implemented within that time period to validate the product as successfully having achieved the agreed upon definition of done.

Scope

The scope of the AALS validation plan reflects only those functionalities outlined in the Functional Requirements Specification.

Description of the solution

A web-based microcontroller is installed on an Arduino Due board and is supported by a SLIMS backend installed on a WIndows Server 2019 VM. The solution is web accessible.

Validation strategy

An agile validation approach will be implemented to include the following practices:

User Requirements Specification document (URS) will be collected on the basis of the intended user and environs.

Functional Requirements Specification document (FRS) will be condensed from the URS.

Performance of an Installation Qualification (IQ) to confirm that the AALS meets functional requirements specifications.

Performance of an Operational Qualification (OQ) to confirm that the AALS meets user requirements specifications.

Upon completion of the IQ and OQ, documentation reflects the outcomes of these validation steps.

User Acceptance Testing Document (UAT) outlines a series of test scripts intended to fulfill a performance qualification or PQ to serve as a verification of reliability in situ.

Validation summary organizes comments, signatures and testing outcomes for the implementation of the Validation plan.

List of required documents

Abbreviation	Description	Owner	Remarks

IQ	Installation Qualification	Sabetpour Wingate Technologies	
OQ	Operational Qualification	Sabetpour Wingate Technologies	
URS	User Requirements Specification	Sabetpour Wingate Technologies	
FRS	Functional Requirements Specification	Sabetpour Wingate Technologies	
CS	Configuration Specification	Sabetpour Wingate Technologies	
UAT	User Acceptance Testing	Sabetpour Wingate Technologies	Also functions as Performance Qualification (PQ)
VP	Validation Plan	Sabetpour Wingate Technologies	(this document)

Validation Summary

Document	**Document ID**	**Signed by**	**Signed on (date)**	**Observed discrepancies**	**Resolution status**

IQ					
OQ					
URS					
FRS					
UAT/PQ					

VP					

Signature box

Signature of this document and documents referred in the Validation Summary completes validation.

Signed on	
Signed by	
Signature	